Md. Uodoy Hossan Rafi

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ACADEMIC PROFILE

Computer Science and Engineering graduate from Daffodil International University (CGPA 3.56/4.00), with experience in distributed systems, software testing and applied machine learning. Built a distributed online judge with Go, C, RabbitMQ and PostgreSQL; research experience includes comparing machine-learning models for Bangladeshi e-commerce sentiment analysis. Solved 3,000+ algorithmic problems and contributed as a contest problem setter and programming mentor. Seeking graduate study in Applied Computing, with interests in software reliability and program analysis.

EDUCATION

B.Sc. in Computer Science and Engineering 01/2022 – 12/2025
Daffodil International University
- CGPA: 3.56/4.00

PUBLICATIONS

Predicting Customer Sentiment from Bangladeshi E-Commerce using Machine Learning Techniques
2025 International Conference on Electrical, Computer and Communication Engineering (ECCE), IEEE, pp. 1–6, 2025.
A. A. Noman, M. D. Uodoy Hossan Rafi, P. S. Antara, M. B. Mahir Tanzim, M. W. Islam Payel and M. Hasan Imam Bijoy.
DOI: 10.1109/ECCE64574.2025.11013335
Compared machine-learning models for sentiment classification using 6,964 Bangladeshi e-commerce reviews; the study reported 94% accuracy with Random Forest.

PROJECTS

Online Judge System (judgenoto)
Go, C, RabbitMQ, PostgreSQL
- Designed and implemented a **distributed online judge** for automated program evaluation
- Built secure execution and judging pipelines for **C and Go** submissions
- Implemented **asynchronous job processing** using RabbitMQ to support concurrent submissions
- Designed PostgreSQL schemas for users, submissions, test cases, and verdicts
- Developed automated **testing, output validation, and verdict classification** mechanisms
- Debugged runtime errors, incorrect outputs, timeouts, and edge-case failures in sandboxed environments

RESEARCH INTERESTS

Automated software testing and debugging; program analysis; software reliability and verification; learning-assisted software engineering.

TECHNICAL SKILLS

Algorithms & Analysis
Data Structures, Graph Algorithms, Dynamic Programming, Complexity Analysis

Software Engineering
Debugging, Edge-Case Analysis, Object-Oriented Programming

Programming Languages
C, C++, Go, Python, Java

Systems & Tools
Git, PostgreSQL, RabbitMQ, VS Code, Sublime Text

SOFTWARE TESTING & DEBUGGING EXPERIENCE

- Automated correctness checking using visible and hidden test cases
- Detection and classification of **Wrong Answer, Runtime Error, Time Limit Exceeded**, and compilation failures
- Systematic debugging of program execution and judging pipelines
- Validation of correctness under adversarial and boundary inputs

COMPETITIVE PROGRAMMING & PROGRAM ANALYSIS

Solved **3,000+ algorithmic problems**, emphasizing correctness, performance optimization, and debugging.

- Codeforces : 1380+ (green)
- AtCoder : 400+ (green)
- LightOJ : 70+
- SPOJ: 50+
- LeetCode : 30+
- Others : 1530+

PROGRAMMING CONTESTS

ICPC Asia Dhaka Regional 2024
—*Rank 84 / 304 teams*

DUET Inter-University Programming Contest 2025
—*Rank 49*

UIU Inter-University Programming Contest 2025
— *Rank 58*

AUST Inter-University Programming Contest 2025
— *Rank 90*

Uttara University IUPC 2025
— *Rank 83*

ACADEMIC & LEADERSHIP ACTIVITIES

Problem Setter and Judge (2023–2025)

- Designed algorithmic problems for university-level programming contests
- Created and validated test cases, verified solutions, and calibrated problem difficulty
- Contributed to preliminary and final rounds of multiple DIU programming contests

Trainer and Mentor — Advanced Programming Camp

- Mentored students in **advanced data structures and algorithms**
- Emphasized debugging strategies, correctness reasoning, and performance analysis

LANGUAGES

- English — Professional proficiency
- Bengali — Native